SAHIL RATHEE

SOFTWARE ENGINEER • BACKEND, DISTRIBUTED SYSTEMS & AGENTIC AI

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TECHNICAL COMPETENCIES

Languages:	Go (Golang), Python, TypeScript, JavaScript, SQL, C++, Dart, Bash
Backend & Frameworks:	Node.js, Express.js, Angular, Next.js (SSR/Edge), FastAPI, React Native (Expo), Flutter, WebSockets
Cloud & DevOps:	Docker, Kubernetes (EKS), AWS (Neptune, OpenSearch, EC2), GCP, Terraform IaC, Ansible, Jenkins CI/CD, Cloudflare
Databases & Caching:	PostgreSQL, MongoDB, Redis, Qdrant Vector DB, Amazon Neptune (Graph DB), NATS JetStream, RabbitMQ
AI & Quantitative:	Agentic Workflows (LangGraph, Function Calling, Multi-Agent RAG), MetaTrader 5, Tick Telemetry, Monte Carlo Simulation

FULL-TIME WORK EXPERIENCE

Software Engineer (Systems & Infrastructure) — Gym Wise India Private Limited

Jul 2024 – Nov 2025 • On-site (Gurugram, India)

- Re-architected monolithic codebase into **3 standalone Flutter/Dart apps** via build flavors: GymWise Customer, GymWise Partner (with biometric attendance), and GymWise Admin, along with a Next.js web portal, live on Google Play and App Store.
- Re-engineered backend microservices from PHP Laravel to **Go (Golang)** backed by Redis buffer and **Qdrant Vector DB** for member discovery and similarity matching.
- Built multi-cloud infrastructure: **AWS** compute and container orchestration managed via **Terraform IaC and Ansible**; GCP for Maps APIs and OAuth services.
- Implemented gym marketplace business logic: zero upfront cost to gym owners, automated commission collection on customer bookings, and algorithmic dynamic pricing based on user engagement.

Tech Stack: *Flutter, Dart, Go (Golang), AWS, GCP, Redis, Qdrant Vector DB, PostgreSQL, Terraform IaC, Ansible, Docker Swarm*

Full Stack Engineer — BigHit

Nov 2022 – Mar 2024 • On-site (Mumbai, India)

- Developed real-time **Socket.io Node.js server** powering dynamic chat for 2,000 Concurrent Active Users (~400 events/sec peak throughput).
- Built real-time score updates and live telemetry engine for live-streamed football tournaments (BigHit Sportz), featuring a dedicated third-umpire web console with role-based access control.
- Migrated legacy relational SQL database to **Amazon Neptune graph DB (Gremlin)**, optimizing multi-degree athlete-to-scout social graph traversal and talent discovery queries.
- Modernized mobile & backend infrastructure, achieving an ~80% reduction in app bundle size and ~60% increase in server efficiency.
- Integrated and managed AWS cloud infrastructure with Amazon Neptune and **OpenSearch** clusters, reducing API response latency under peak tournament load.

Tech Stack: *TypeScript, React Native, Node.js, Socket.io, GraphQL, AWS Neptune (Graph DB), OpenSearch, Docker, Kubernetes, AWS IaC, CI/CD*

Full Stack Developer — Gamix Labs

Mar 2022 – Oct 2022 • On-site (Noida, India)

- Engineered real-time mathematical simulation engines for Vegas slot machines (HTML5 / Facebook Instant Games on Cocos Creator Engine) with Node.js and RabbitMQ event streaming backends.
- Formulated and verified Return-to-Player (RTP) and win frequency probability algorithms via Monte Carlo simulations (10M+ spins at 99.9% confidence interval) ensuring statistical house edge while maintaining high user engagement.
- Architected multi-database backends across client games utilizing both SQL and NoSQL engines (MongoDB and Firebase Realtime Database / Firestore).
- Collaborated with gaming clients (OneFC, Bombay Play) to deliver responsive PWA online gaming platforms and HTML5 titles with Redis and MongoDB caching.

Tech Stack: *TypeScript, Cocos Creator Engine, Node.js, RabbitMQ, MongoDB, Firebase, SQL, Redis, Angular, Flutter, GCP*

FREELANCE & CLIENT PROJECTS

Full Stack Software Engineer — NaiBooking

Aug 2026 – Present • Remote (Bangalore, India)

- Architected full-stack salon & barber SaaS marketplace: web portal and admin console in **React / Next.js**, client and business mobile apps in **React Native (Expo)** live on Google Play & App Store.
- Engineered Node.js backend leveraging **MongoDB** and **PostgreSQL** with transaction ledgers, advisory locking, and multi-tier Redis caching to handle high-concurrency booking spikes.
- Automated AWS cloud infrastructure via **Terraform IaC**; integrated Razorpay automated subscription billing for salon CRM with an architectural roadmap for global payment expansion.

Tech Stack: *TypeScript, React, Next.js, React Native (Expo), Node.js, MongoDB, PostgreSQL, Redis, AWS, Terraform IaC, Docker*

Full Stack Engineer — GrowthHikes

2026 • Remote

- Engineered modern web platform and interactive client calculators using Next.js/React deployed serverless to Cloudflare Workers.
- Built automated agentic content workflows and scheduled marketing operations pipelines with real-time analytics aggregation.

Tech Stack: *React, Next.js, Cloudflare Workers, Node.js, Python, Analytics APIs*

Software Engineer (Contract) — Orealfit May 2026 – Jul 2026 • On-site (Sector 44, Gurugram, India)

- Consolidated sports simulation services into a unified monorepo orchestrated with Docker Compose for local development and simulation.
- Built internal billing and GST invoicing engine for Indian merchants on Next.js 14 and Drizzle, deployed serverless to Cloudflare Pages.

Tech Stack: *Flutter, Next.js, Go, Node.js, PostgreSQL, Supabase, Cloudflare Workers, Docker*

PERSONAL & OPEN-SOURCE PROJECTS

Meeseeks — Autonomous Multi-Agent AI Framework meeseeks.infortts.site

- Built an autonomous multi-agent orchestration framework supporting tool-calling supervisor loops, vector-backed shared memory, and self-healing diagnostic routines.
- Engineered AST context compression and isolated container execution environments for reliable agentic task completion.

Tech Stack: *Python, Go, FastAPI, LangGraph, Docker, Qdrant Vector DB, Redis*

Mitochondria & Zenith — Quantitative Telemetry & Execution Workstation Quantitative Engineering

- Architected an algorithmic trading engine with MetaTrader 5 execution bridge, tick-level market telemetry streaming, and vectorized backtesting.
- Engineered low-latency in-memory ring buffers and real-time event streaming via NATS JetStream with automated drawdown and position-sizing safeguards.

Tech Stack: *Python, C++, Go, NATS JetStream, PostgreSQL, Qdrant, MetaTrader 5, Flutter*

EDUCATION

Bachelor of Science (B.Sc. Hons) in Computer Science — DDUC, University of Delhi 2018 – 2021 • New Delhi, India